

- Project Title: Smart Blood Inventory & Donor Engagement Platform
- Domain: HT-07 (Health-Tech)
- Team Name: Health X Innovators
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- Department: CSE
- *Event: Udbhava X'26, Coding Club SR University*



Problem Scenario

The Real-World Situation:

A healthcare system functions like a life-support network where timely access to blood is critical for surgeries, emergencies, and treatments. Hospitals, blood banks, and donors are interconnected stakeholders who rely on efficient coordination to ensure availability. However, due to the absence of a centralized system, blood banks often operate in isolation, making it difficult to quickly locate required or rare blood types. This lack of coordination leads to delays, wastage of blood units, and increased risk to patient lives during critical situations where every second matters.

The Current Failure:

When blood is required, the process is largely manual and informal—relying on phone calls to other hospitals or urgent posts on social media to find donors. This creates a chaotic and unstructured system where requests are not properly tracked and often delayed. Due to the absence of a centralized platform, hospitals struggle to verify availability in real time, and critical requests may go unnoticed. This results in wasted time, expired blood units, and increased risk to patients during emergencies.



Current Pain Points (Why it Breaks)

- **No Centralized "Source of Truth":** Blood inventory data is scattered across individual hospitals, making it impossible to get real-time availability across the network.
- **The "Duplicate Effort" Problem:** Multiple hospitals simultaneously search for the same blood type through calls or social media, wasting critical time during emergencies.
- **Inability to Prioritize Emergencies:** Critical cases (e.g., accident patients needing rare blood) are not distinguished from routine requests, causing dangerous delays.
- **Blood Wastage Due to Poor Tracking:** Without proper expiry monitoring and FIFO usage, many blood units expire unused while shortages exist elsewhere.
- **Lack of Donor Engagement:** Donors are not notified when their specific blood type is urgently needed, leading to missed opportunities to save lives.
- **No Historical Data for Planning:** There are no proper records to analyze trends like seasonal shortages or demand for rare blood types, limiting effective decision-making.



Objectives & Success Criteria



Primary Objective:

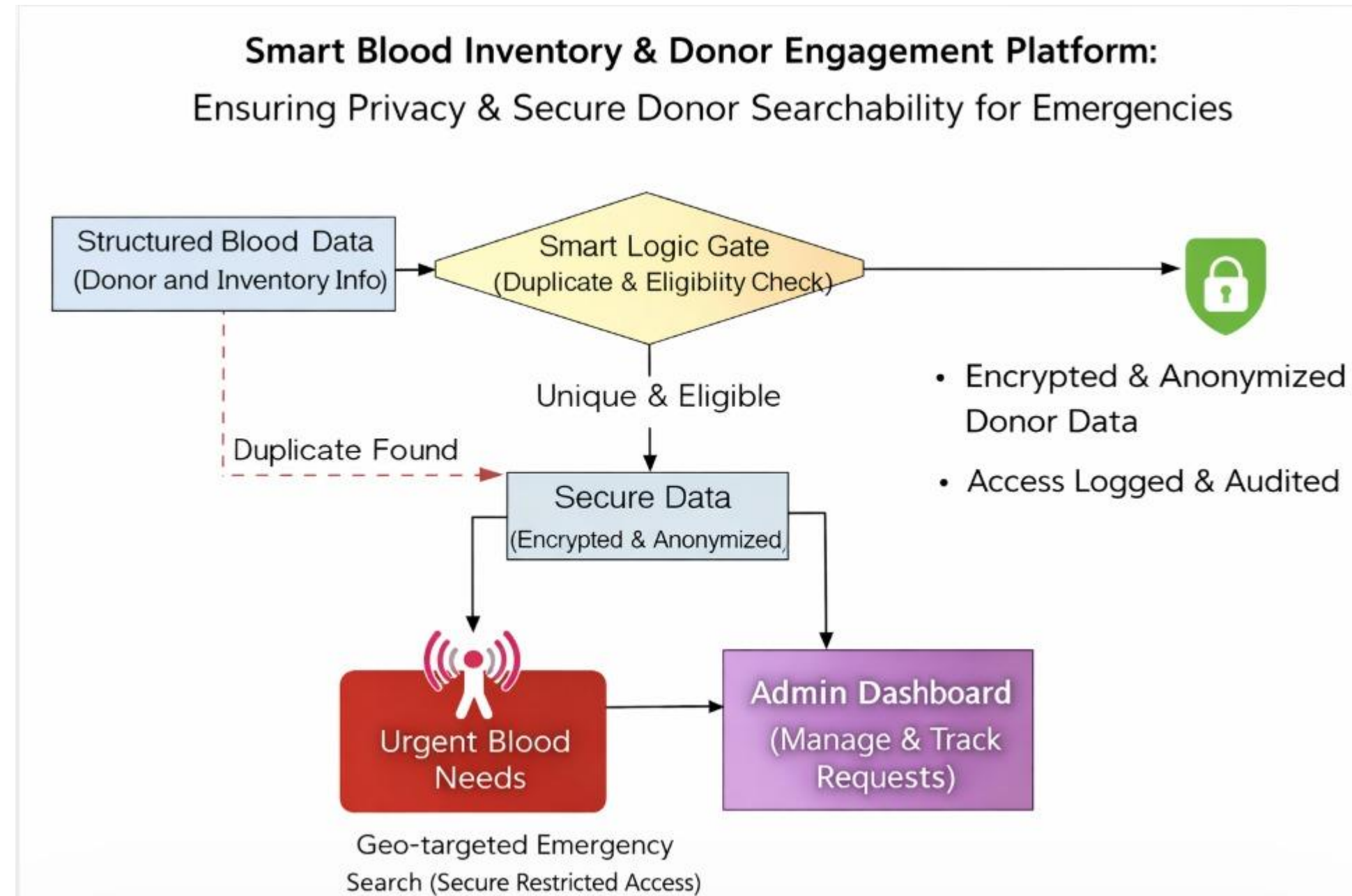
To replace unstructured communication with a centralized, digital platform that standardizes grievance reporting and enforces accountability

. Success Criteria:

- Structured tracking of blood inventory and donor data
- Smart matching of blood units and eligible donors
- Real-time status updates for requests
- Quick emergency alerts for rare blood types
- Reduced blood wastage using expiry tracking (FIFO)
- Safe donation control (prevent over-donation)



Proposed Solution Strategy



Key Solution Strategies:

1. Input Standardization:

Ensure structured data entry for blood units and donors (blood type, location, expiry, availability) to avoid confusion and enable accurate tracking.

2. Role-Based Interfaces:

Provide separate dashboards for:

- **Hospitals/Blood Banks** → manage inventory & requests
- **Donors** → view requests, check eligibility, respond to alerts

3. Smart Matching Logic:

Automatically match blood requests with available units or nearby eligible donors based on blood type, location, and urgency.

4. Emergency Broadcast System:

Instantly notify relevant donors during urgent or rare blood requirements to reduce response time.

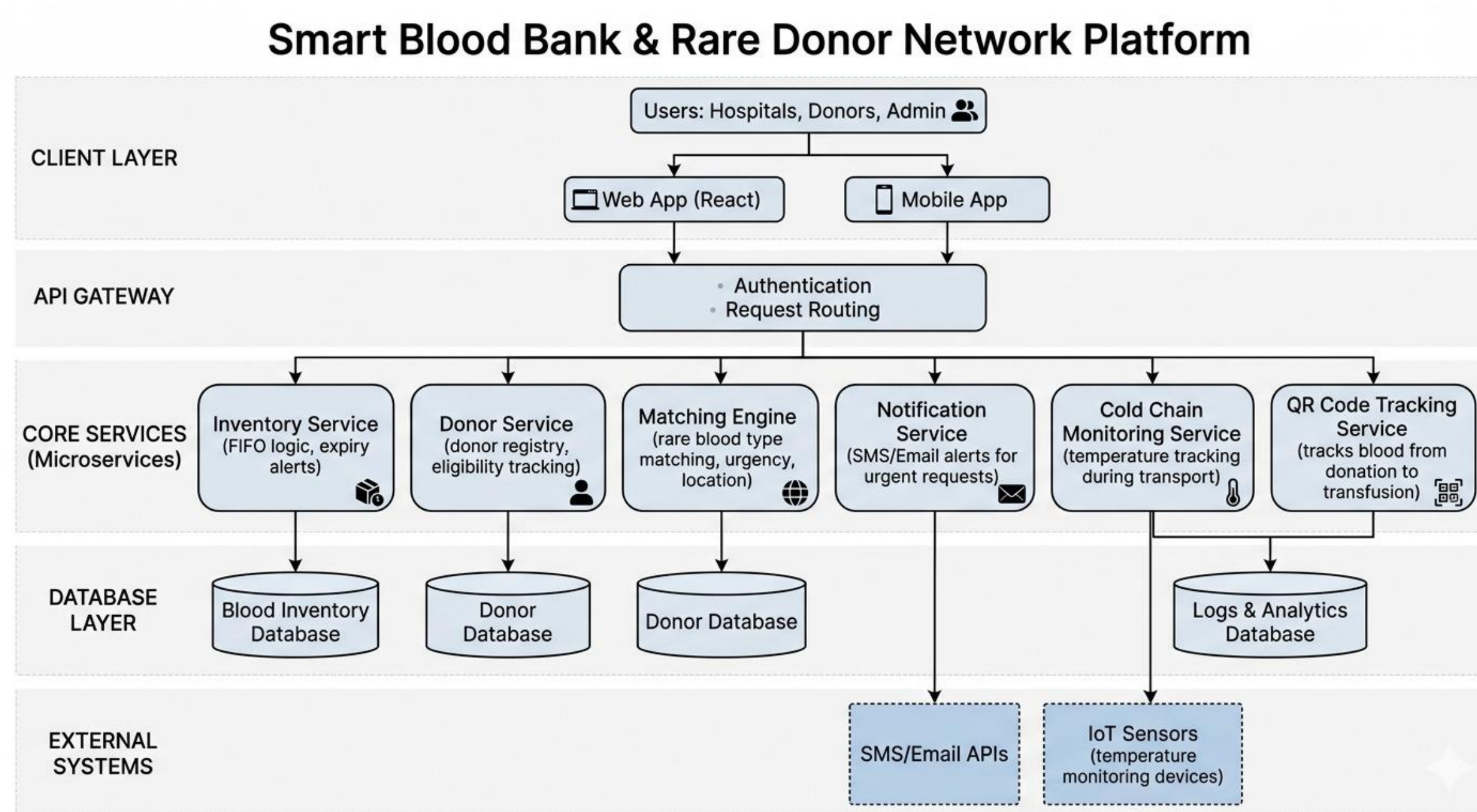
5. Inventory Optimization (FIFO Logic):

Prioritize usage of older blood units first and generate alerts before expiry to minimize wastage.



System Architecture

Architectural Overview (3-Tier):



Frontend Layer (React):

Responsive dashboards for **Hospitals/Blood Banks** (inventory management, requests) and **Donors** (alerts, availability, eligibility status).

Backend Logic Layer:

Handles **data validation, authentication**, and executes core logic such as **blood unit matching, donor eligibility checks, FIFO inventory management, and emergency alert broadcasting**.

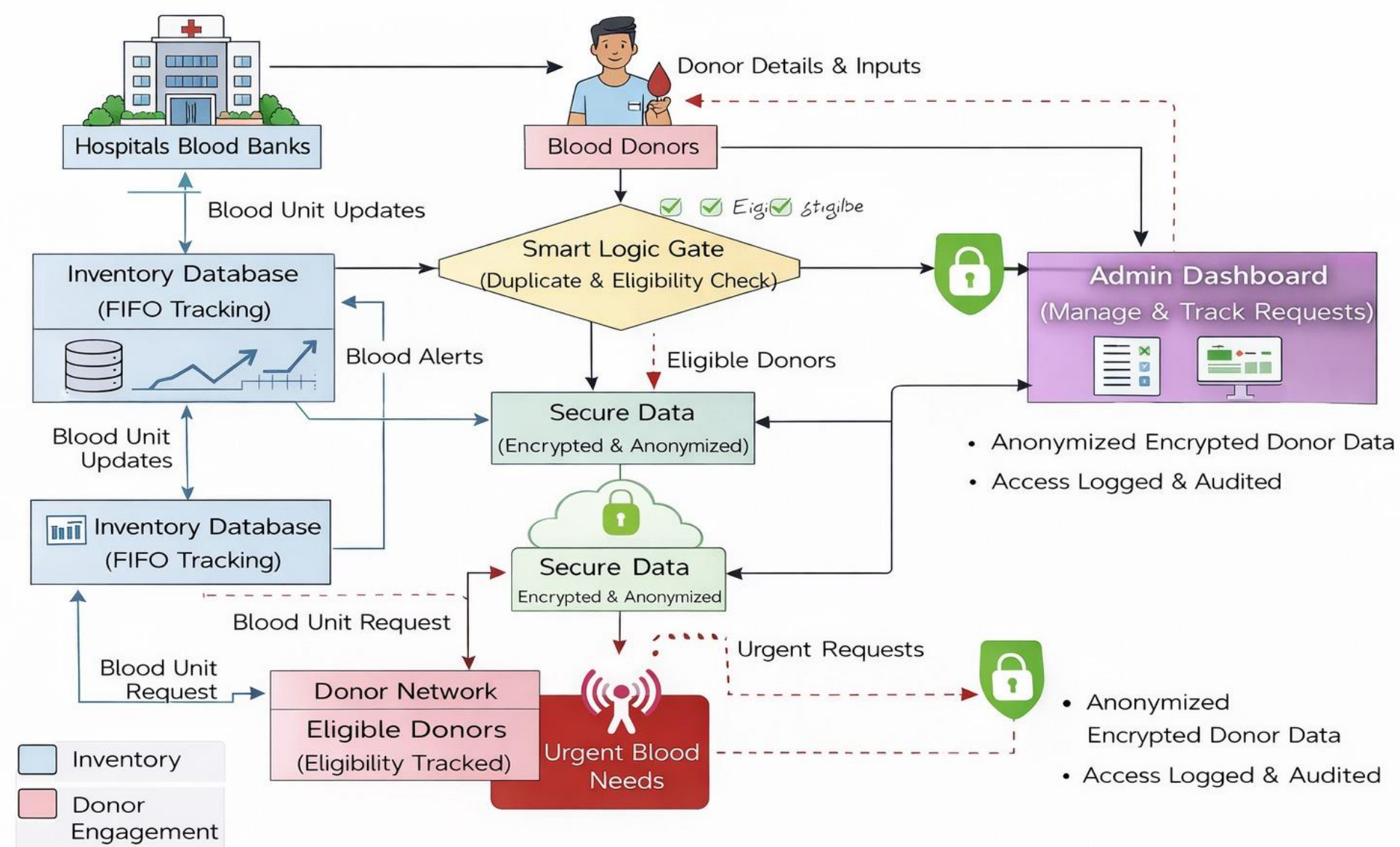
Data Persistence Layer:

Stores structured relationships between **blood units, donors, hospitals, requests, and transactions** for reliable tracking, retrieval, and analysis.



Data Flow / Workflow

Smart Blood Inventory & Donor Engagement Platform:
Data Flow & Workflow



1. Request Submission:

Hospital submits a blood request by selecting **blood type, location, and urgency level** OR a new blood unit is added after donation.

2. Validation & Matching Logic:

System checks available inventory and donor database.

Decision:

- If blood unit is available → **Reserve & Proceed**
 - If not available → **Trigger donor matching & alert system**
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3. Smart Matching & Alerts:

Nearby eligible donors are identified and notified.

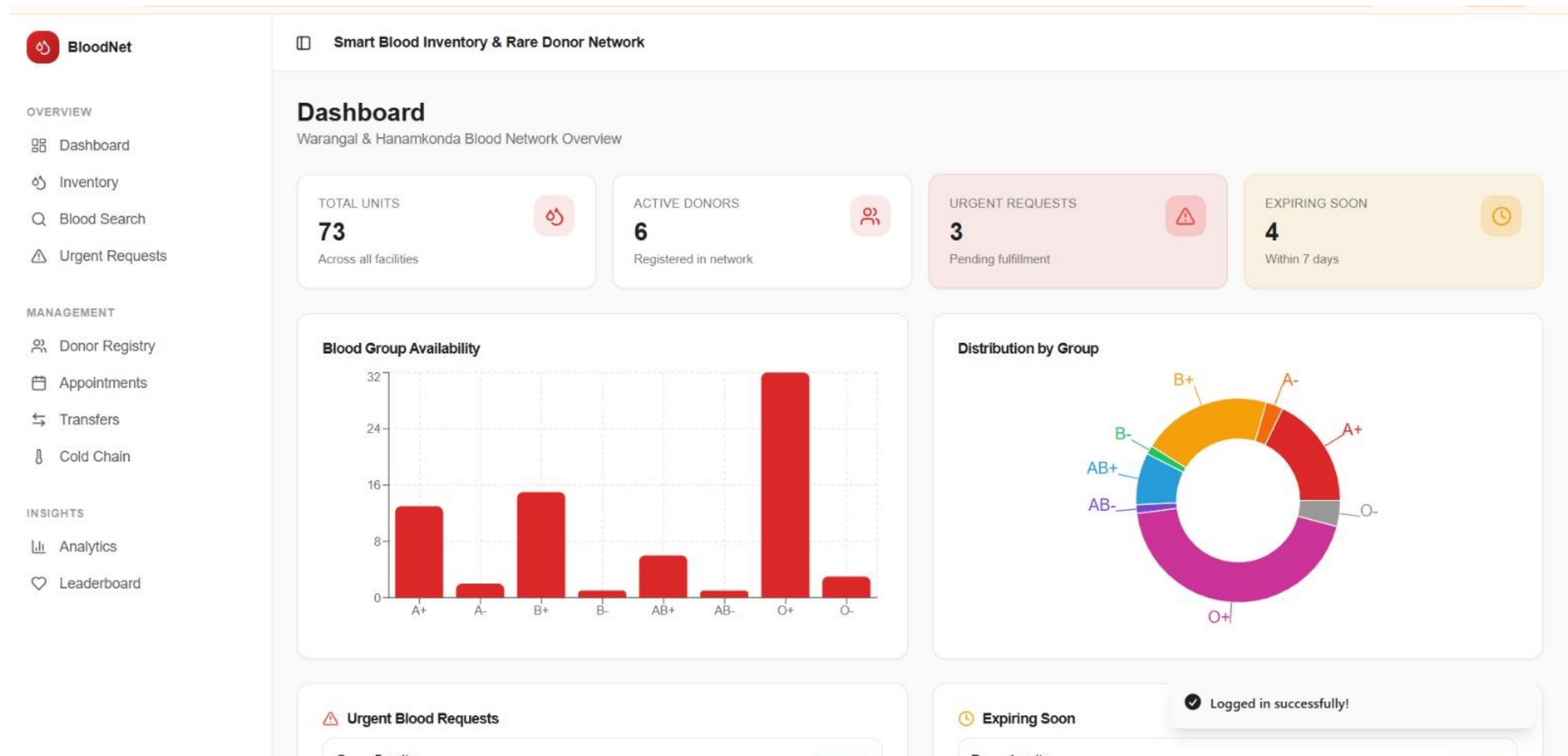
Hospitals can view matched results in real-time, with **urgent requests highlighted (e.g., Red)**.

4. Fulfillment & Tracking:

Once blood is arranged (from inventory or donor), status is updated to **"Fulfilled"**, and the entire process is tracked from request to delivery.



Implementation & Demo Reference



Implemented Features (Reference Prototype):

- Role-based login (Hospital/Donor)
- Structured request & donor forms
- Real-time inventory dashboard
- Smart donor matching
- Emergency alert system
- Priority highlighting (urgent cases)
- Status tracking (Requested → Fulfilled)

LINK REQUIREMENT:

GitHub Repo:

https://github.com/jupakakeerthipriya/HealthXinnovators_.Blood-Bank-Rare-Donor-Network

Live Demo: <https://rare-drop-connect.lovable.app>



Impact & Benefits

For Patients:

Faster access to required blood, especially during emergencies
Reduced risk due to delays in finding rare blood types

For Hospitals/Blood Banks:

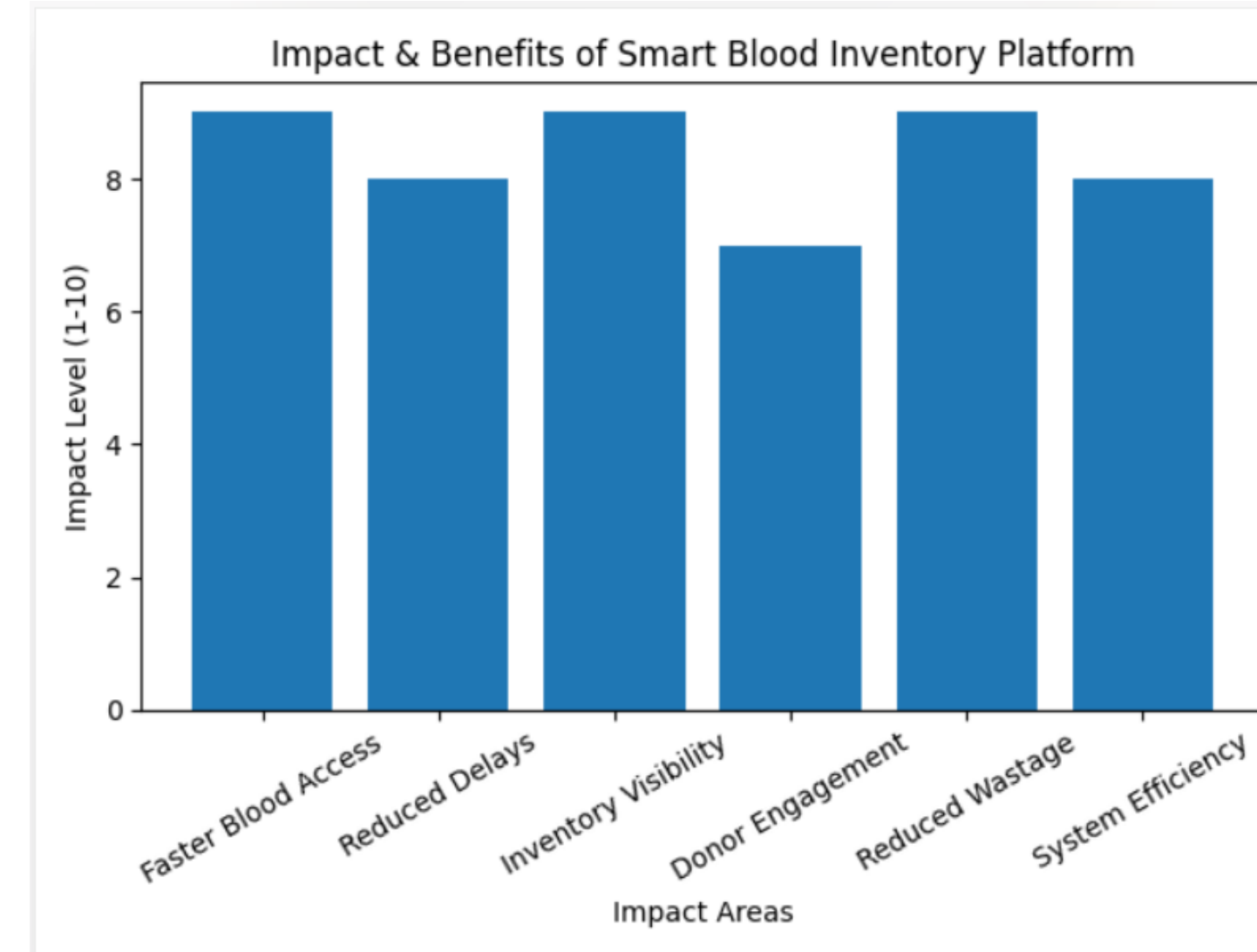
Real-time visibility of blood inventory across networks
Reduced time spent searching for blood units

For Donors:

Timely notifications when their blood type is needed
Increased engagement and contribution to saving lives

For the Healthcare System:

Reduced blood wastage through proper tracking (FIFO)
Access to historical data for better planning and demand forecasting
More efficient, reliable, and life-saving operations



Prototype Data Persistence:

For this hackathon prototype, data may be stored using simple/local storage or a lightweight database. A production system would require a secure, scalable cloud database for real-time multi-hospital access.

Authentication:

The current system uses basic login functionality and does not yet integrate with official hospital systems or secure health identity frameworks.

Cold-Chain Integration:

Real-time temperature tracking is simulated; actual implementation would require IoT devices for accurate monitoring during storage and transport.

Offline Capability:

The system depends on internet connectivity and does not yet support full offline functionality with later synchronization.



Future Scope & Roadmap

Phase 2: Intelligence & Mobile

AI-Driven Demand Prediction:

Use AI/ML to predict blood demand based on historical data, seasonal trends, and emergencies.

Smart Donor Matching (AI):

Enhance matching using location, past donation behavior, and response patterns.

Mobile App Development:

Build a Flutter/React Native app with **real-time alerts, GPS-based donor tracking, and instant response features.**

Phase 3: Smart Integration

IoT Cold-Chain Integration:

Connect with smart sensors to automatically monitor and log blood temperature during storage and transport.

Healthcare System Integration:

Integrate with hospital databases and government health systems for seamless data sharing.

Automated Alert System:

Trigger alerts automatically when blood stock goes below critical levels.



Development Roadmap: Smart Blood Inventory & Donor Engagement Platform

